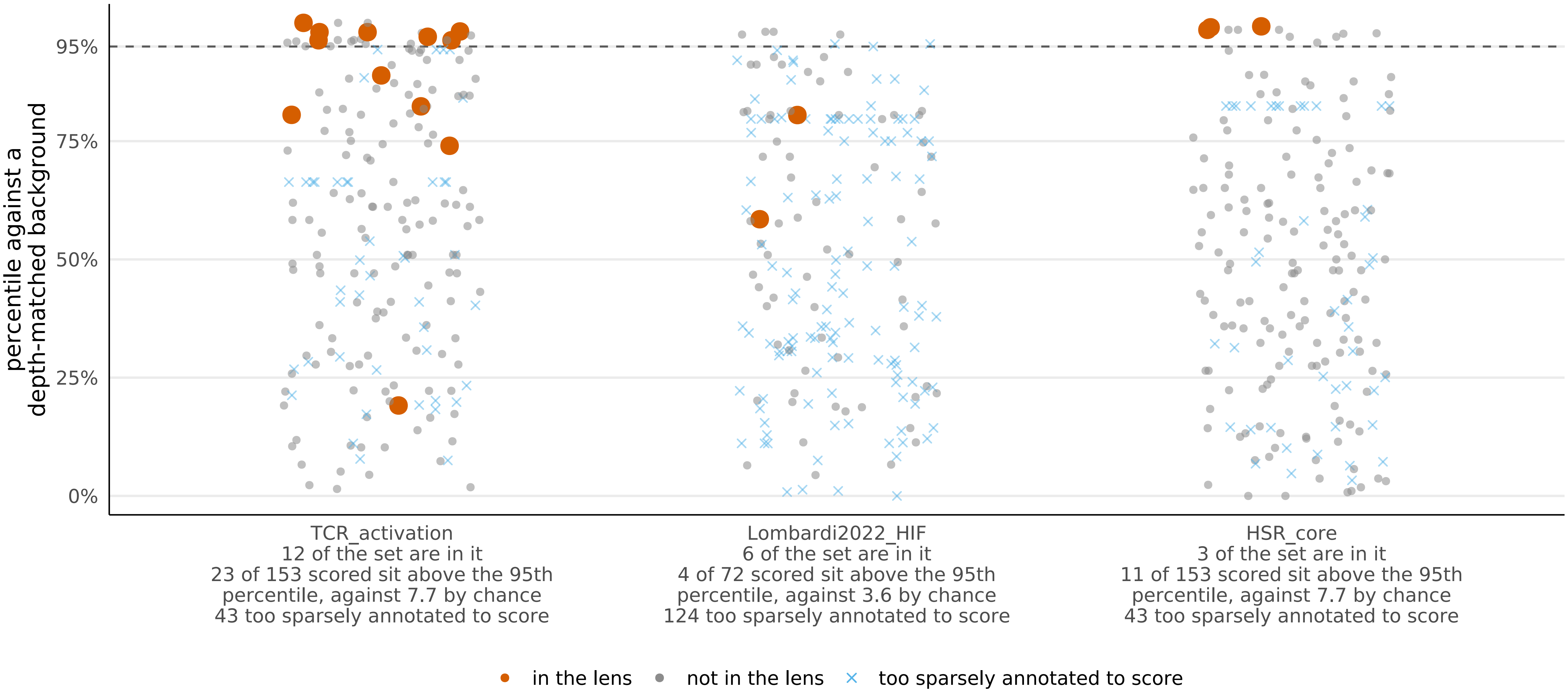
Membership is binary, and the genes it leaves out are not all equally far away

The curated lenses contain 21 of the 196 genes between them. This asks the graded question of the rest: how close does each gene's GO BP annotation sit to each lens, measured against background genes annotated to a similar number of terms. A gene scoring high here resembles the lens in what has been recorded about it. That is not evidence it participates in the process the lens is named for.

For the genes no lens contains, how close does each one sit?

Each grey dot is one gene of the set, placed by how its annotation compares with the nearest OTHER member of that lens, scored against background genes carrying a similar number of terms. Red dots are genes the lens already contains, and they act as the control: a working measure puts them high without being shown the answer. Crosses are genes too sparsely annotated to rank within their band.



Why the percentiles above are taken within a depth band

A gene annotated to more terms sits closer to every reference set, whatever the set is. That is a property of how much has been recorded about the gene, not of the biology, and pooling the background over all depths would read it as signal.